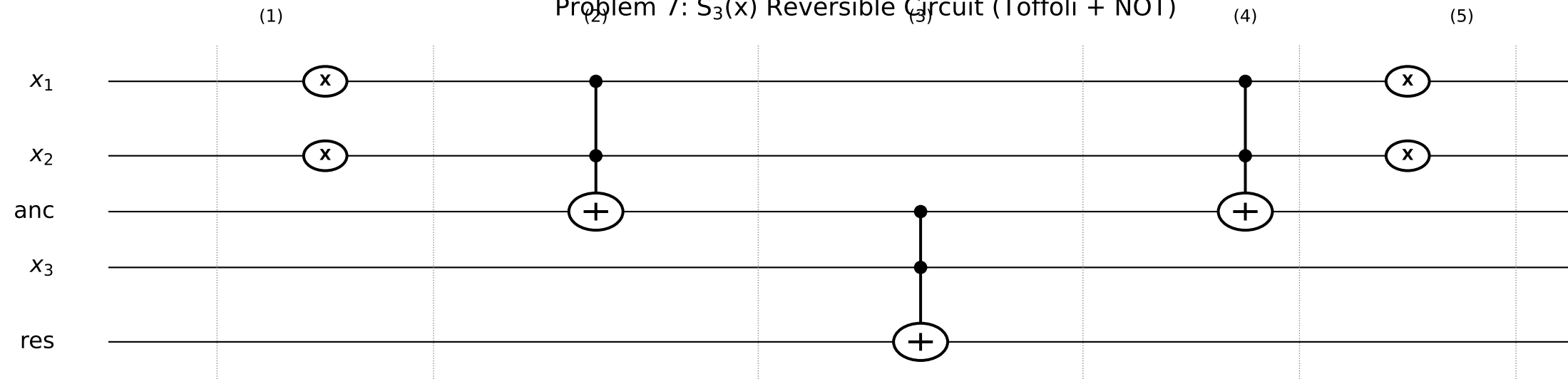


Problem 7: $S_3(x)$ Reversible Circuit (Toffoli + NOT)



$S_3(x) = \neg x_1 \wedge \neg x_2 \wedge x_3$
 Gates: 4 NOT + 3 Toffoli = 7
 Qubits: 3 (input) + 1 (ancilla) + 1 (result) = 5
 Product: 7 x 5 = 35
 Steps: (1) flip x_1, x_2 (2) anc = $\neg x_1 \wedge \neg x_2$
 (3) res = anc $\wedge x_3$ (4) uncompute anc (5) uncompute flip